IT INFRASTRUCTURE AND COMPUTER NETWORKS

Week 9

How many IPv4 addresses we have?

~4 billion

NOT ENOUGH!

Solutions?

IPv6

NAT

IP address control

IP addresses

Private

Local
Non-routable

10.0.0.0/8
172.16.0.0/12
192.168.0.0/16

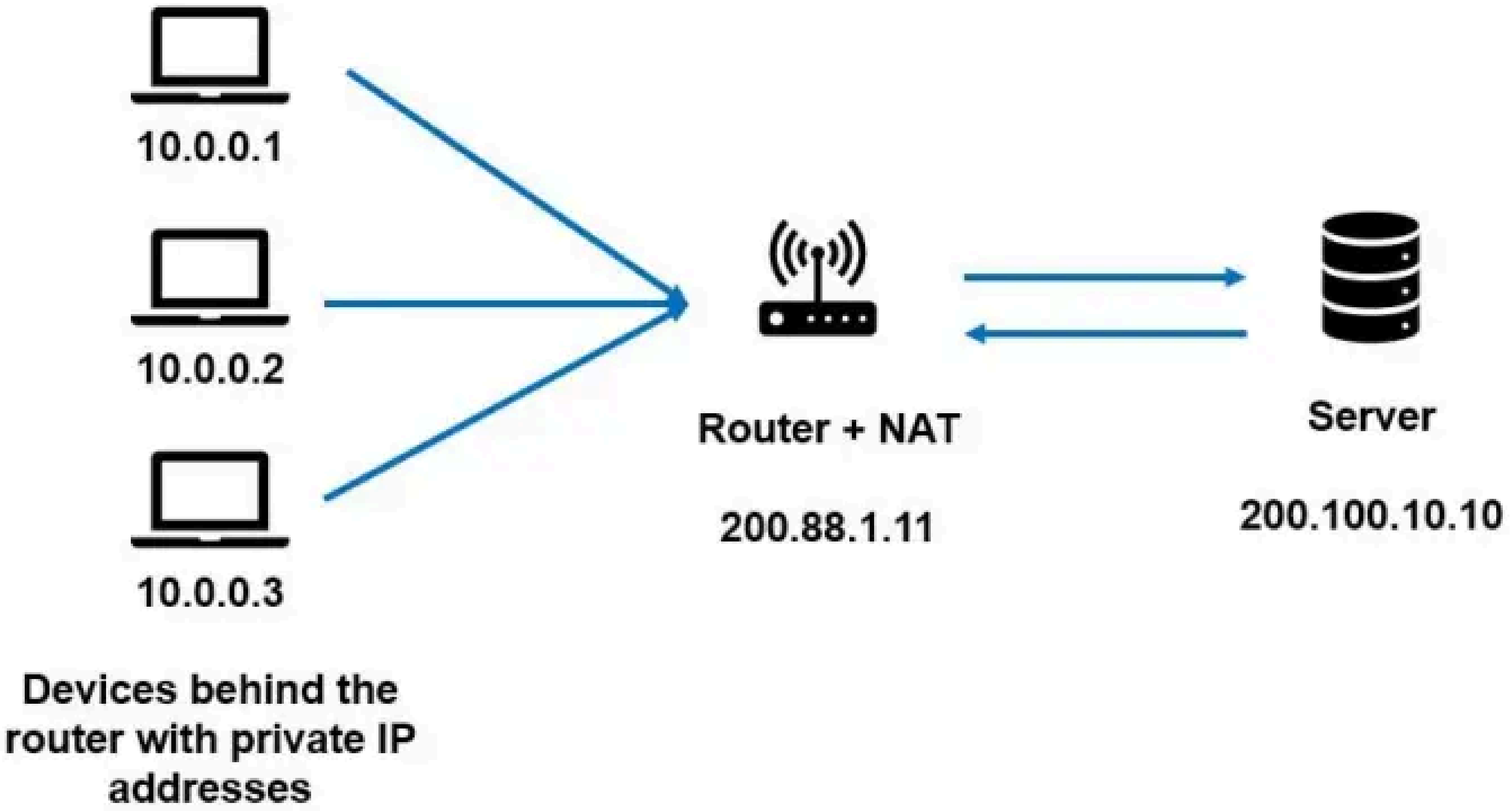
Public

Unique
Given by ISPs

Other IP addresses

NAT

Network Address Translation (NAT) enables private IP networks to use the internet and cloud by translating (internal) private IP addresses to (external) public IP addresses.



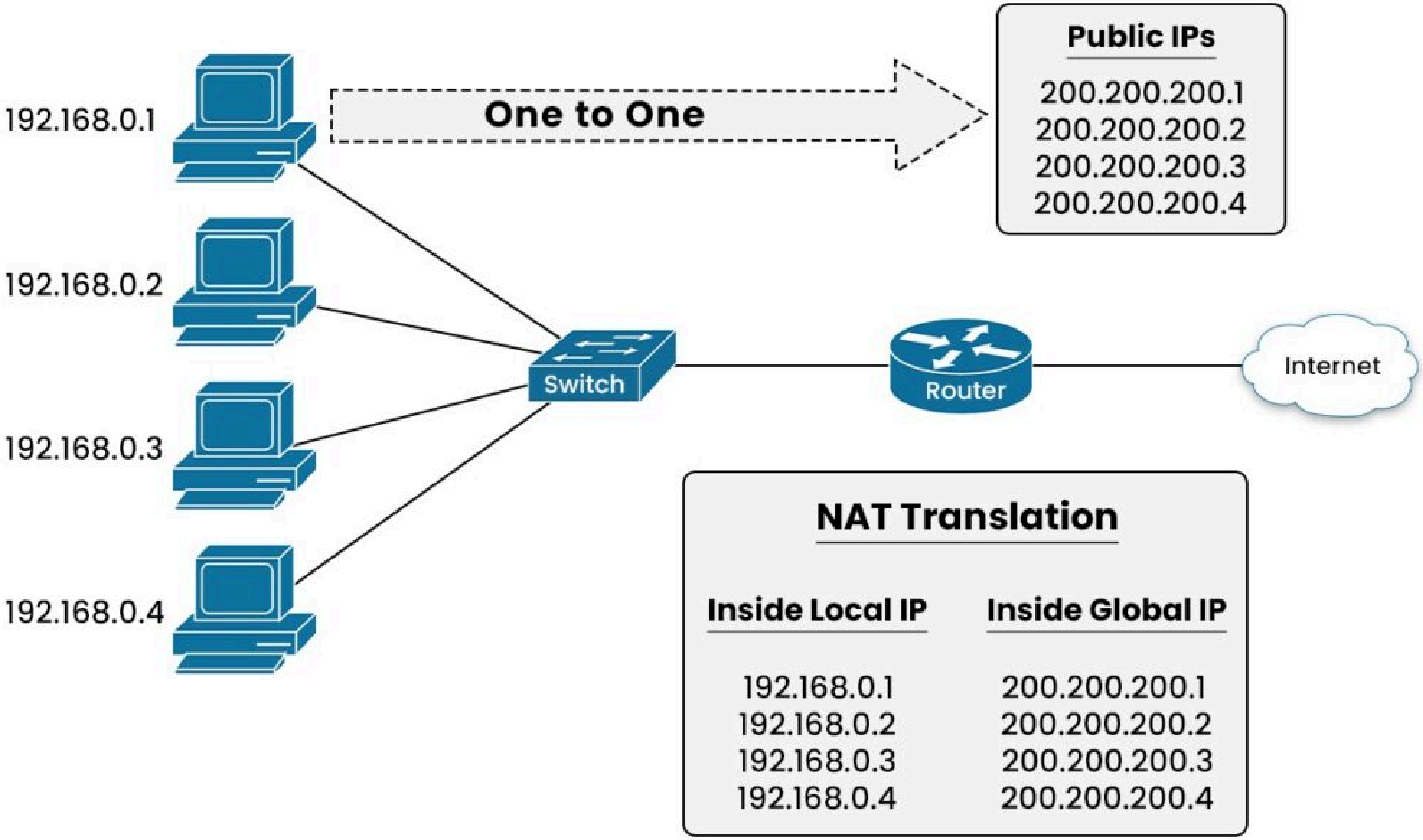
Types of NAT

Static NAT (One-to-One): Maps a single private IP address to a unique public IP address, often used for servers.

Dynamic NAT (Many-to-Many): Maps internal IP addresses to a pool of available public IP addresses.

NAT Overload (Many-to-One): Maps multiple private IP addresses to a single public IP address using different port numbers - also called Port Address Translation (PAT).

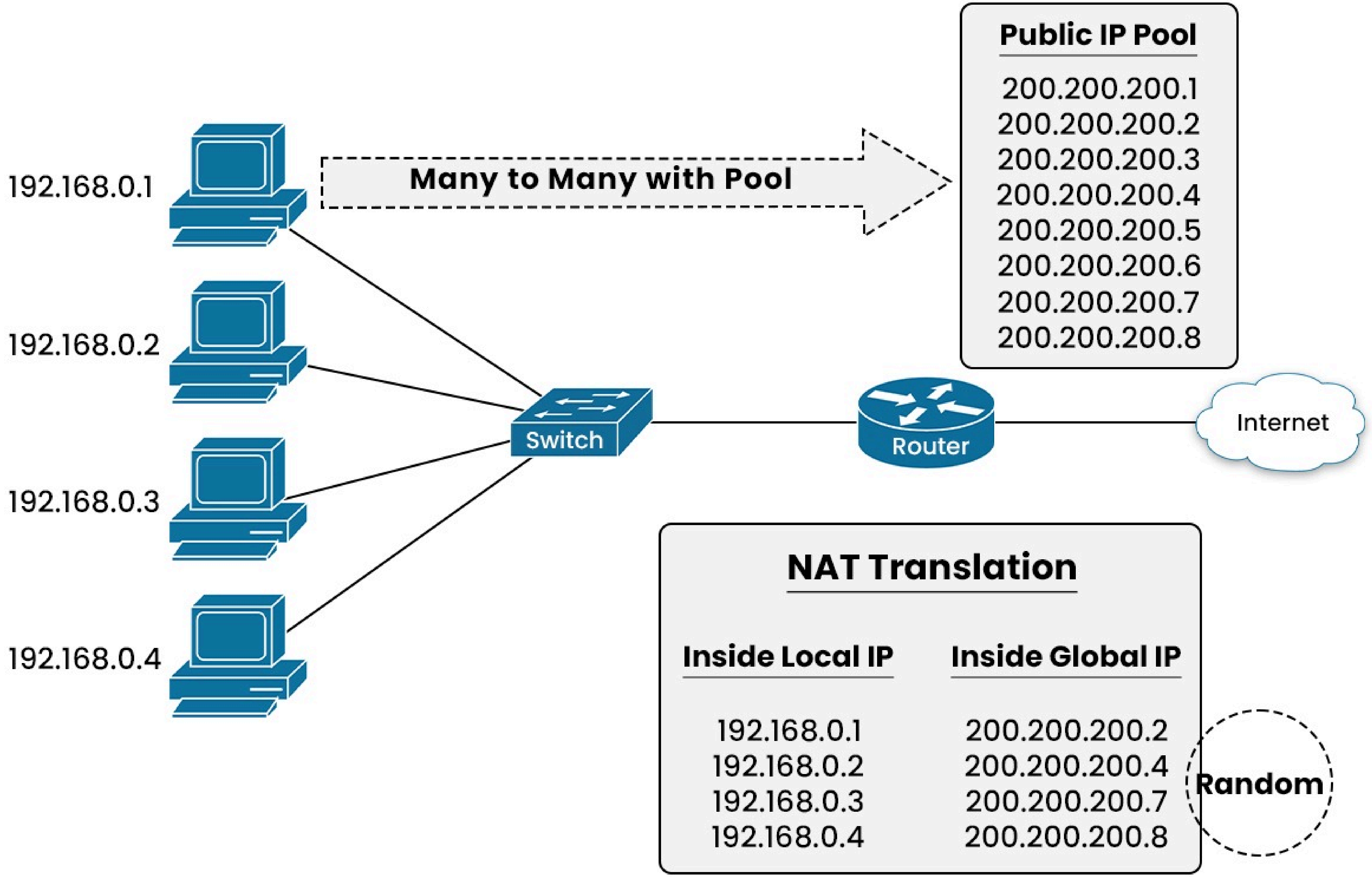
STATIC NAT



NAT Translation

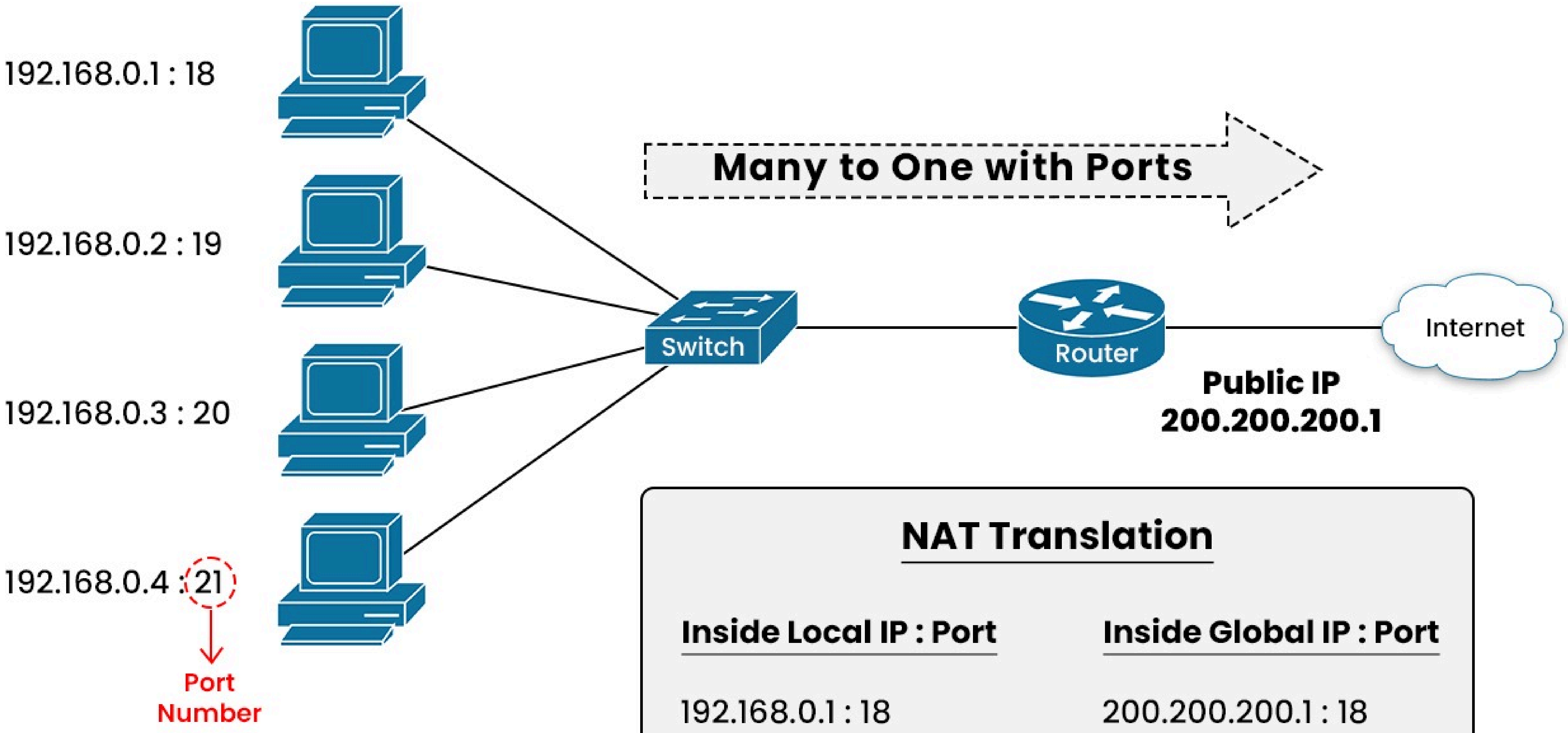
Inside Local IP	Inside Global IP
192.168.0.1	200.200.200.1
192.168.0.2	200.200.200.2
192.168.0.3	200.200.200.3
192.168.0.4	200.200.200.4

DYNAMIC NAT



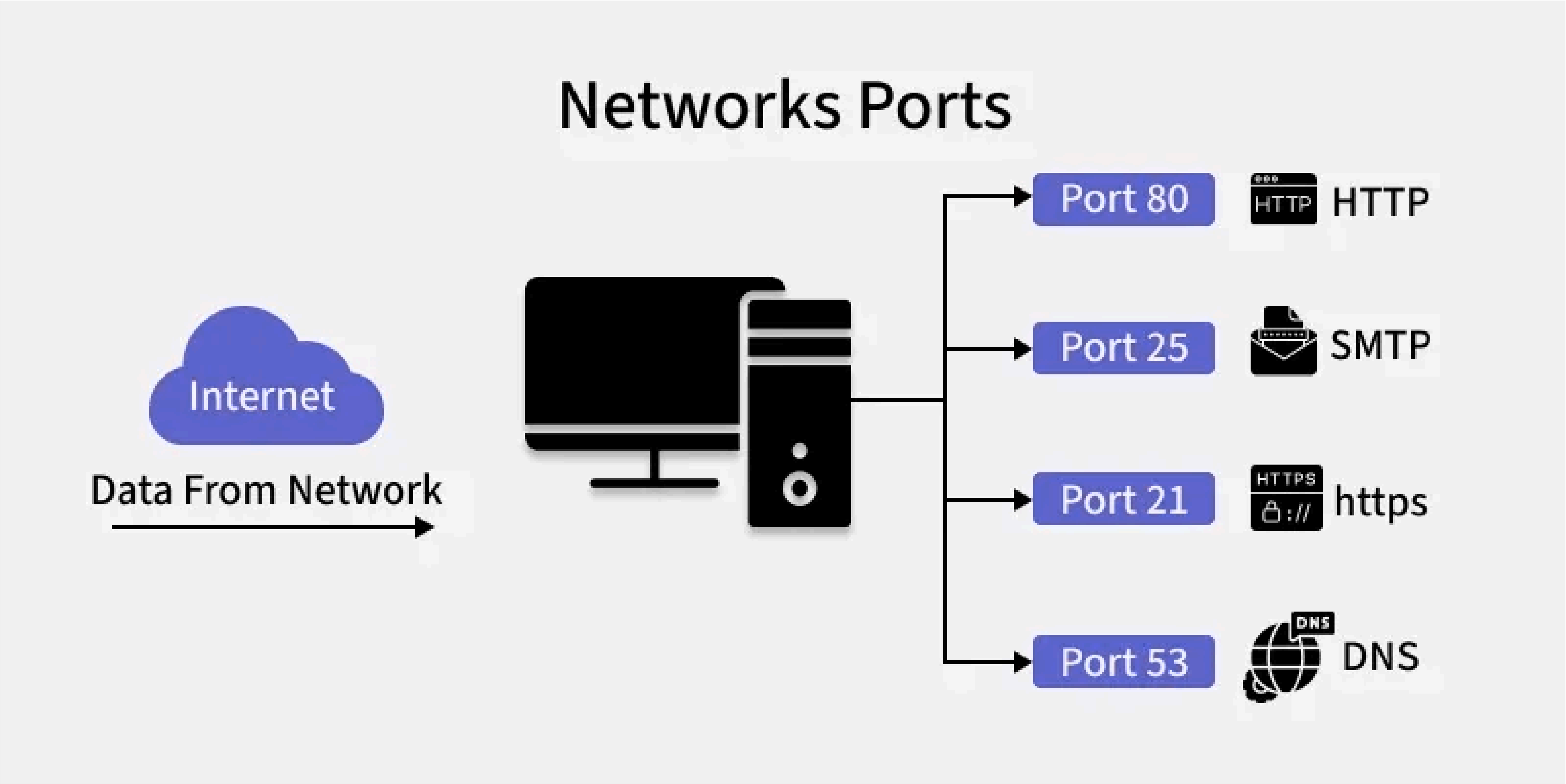
Public IP Pool
200.200.200.1
200.200.200.2
200.200.200.3
200.200.200.4
200.200.200.5
200.200.200.6
200.200.200.7
200.200.200.8
Many to Many with Pool
192.168.0.1
192.168.0.2
192.168.0.3
192.168.0.4
Switch
Router
Internet
NAT Translation
Inside Local IP
Inside Global IP
192.168.0.1
192.168.0.2
192.168.0.3
192.168.0.4
200.200.200.2
200.200.200.4
200.200.200.7
200.200.200.8
Random

PAT (NAT Overload)



NAT Translation

Inside Local IP : Port	Inside Global IP : Port
192.168.0.1 : 18	200.200.200.1 : 18
192.168.0.2 : 19	200.200.200.1 : 19
192.168.0.3 : 20	200.200.200.1 : 20
192.168.0.4 : 21	200.200.200.1 : 21



A port is a logical identifier used to distinguish different applications or services on a device, allowing network traffic to reach the correct program.

Common network ports

Port	Service name	Transport protocol
20, 21	File Transfer Protocol (FTP)	TCP
22	Secure Shell (SSH)	TCP and UDP
23	Telnet	TCP
25	Simple Mail Transfer Protocol (SMTP)	TCP
50, 51	IPSec	
53	Domain Name System (DNS)	TCP and UDP
67, 68	Dynamic Host Configuration Protocol (DHCP)	UDP
69	Trivial File Transfer Protocol (TFTP)	UDP
80	HyperText Transfer Protocol (HTTP)	TCP
110	Post Office Protocol (POP3)	TCP

Any questions?